

Unit-4 Metrics

Introduction

- In software engineering, Metrics are essential for evaluating both the processes involved in software development and the products produced.
- Two primary categories of metrics are **process metrics** and **product metrics**.
- Within these categories, specific metrics such as Lines of Code (LOC), Function Points, and Token Counts are commonly used.

Process Metrics

- Process metrics are used to assess the effectiveness and efficiency of software development processes.
- They help in understanding how well the development team is performing and can guide improvements in methodologies and practices.

Key Characteristics of Process Metrics

- Efficiency Measurement:
 - Evaluate how effectively resources (time, personnel) are utilized during the development process.
- Quality Assessment:
 - Measure the quality of the processes, including defect detection rates and process maturity.
- Improvement Tracking:
 - Monitor changes over time to identify trends and pinpoint areas that need improvement.

Examples of Process Metrics

- 1. **Defect Density**: Number of defects per unit size (e.g., per 1,000 lines of code) to assess quality.
- 2. **Cycle Time**: The time taken to complete a specific task or feature from start to finish.
- 3. **Lead Time**: The total time taken from the initial idea to the delivery of the software product.

Product Metrics

- Product metrics focus on the attributes and quality of the software product itself. They help in evaluating various aspects such as size, complexity, performance, and quality.
- Key Characteristics of Product Metrics
 - - **Quantitative Assessment** : Provide measurable data about the software's characteristics.
 - - **Quality Evaluation** : Help assess how well the product meets user requirements and expectations.
 - - **Complexity Measurement** : Offer insights into how complex or maintainable a software product is.

Examples of Product Metrics

- 1. **Lines of Code (LOC)**:
 - - Measures the total number of lines in a program's source code.
 - - Useful for estimating project size and complexity but can be misleading as it doesn't account for code quality or functionality.

Product Metrics(Contd..)

- 2. **Function Points**:
 - - A standardized method for measuring software size based on its functionality from a user perspective.
 - - Considers inputs, outputs, user interactions, files, and interfaces to provide a more accurate measure of productivity and project scope than LOC.
- 3. **Token Counts**:
 - - Refers to counting distinct elements in the source code (like keywords, operators, identifiers).
 - - Useful for analyzing code complexity and can help in understanding how many unique constructs are present in a program.